

Ice Tasks 3

Research problem

OWASP (2023) argues that with the rapid growth of digital banking in South Africa, there has been an increase of reliance on web-based applications for services and transactions. These systems enhance efficiency and accessibility. These very systems are targeted by cybercriminals which leads to data breaches, exposing confidential customer information and financial loss. Web-based applications still remain endangered to the common security risks e.g. broken authentication and injection attacks, these emphasize the necessity to have secure development practices and this is a global issue.

Although there is an availability of secure software development framework, there are several applications that still have experience vulnerabilities caused by the inadequate integration of security practices throughout the software development lifecycle (SDLC). NIST (2022) states that security must be incorporated into all stages of development because this will reduce risks and improve system resilience. Snyk (2023) states that although the traditional approaches have frequently treated security as a separate phase and this has led to having vulnerabilities being identified when it is too late and that increases the costs and complexity of remediation.

South African banking sectors' dependance on digital platforms has increased which has made financial institutions major targets to cybercrimes. SABRIC (2024) states that digital banking fraud still continues to rise which poses significant risks to both the financial institutions and their customers. This shines a spotlight on the necessity of having more effective integration of security practices into the SDLC for banking web-based applications in South Africa.

Relevance/motivation

Globally, integration of security practices into SDLC has become a priority in software engineering especially in industries that deal with sensitive data such as banking. OWASP (2023) argues that Secure SDLC and DevSecOps frameworks emphasize incorporating security throughout all the stages of development because that ensures that all vulnerabilities are identified and get addressed early. IBM (2023) states that this approach doesn't just improve the application's security it also reduces the cost and impact of potential security breaches.

A lot of the existing research focus on global frameworks and best practices with limited attention given to their implementation within the specific national contexts. Nationally, there is an increase of incidence of cybercrime and digital banking fraud which shines a spotlight on the urgent necessity of having strong applications security measures. SABRIC (2024) reports that financial institutions continue to face significant losses because cyber-related crimes.

NIST (2022) and OWASP (2023) argue that there is limited research going through how South African banking institutions integrate security practices into the SDLC and the challenges they face during the process. This a relevant study because it seeks to address this gap by analysing the implementation of secure SDLC practices within the South African banking sector and the findings may help by contributing to improved security strategies, increasing trust in digital banking systems and enhancing protection of confidential financial data.

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